Lesson notes · for you

Make it automatic: unit check

IoT · Level 3 · Unit check

Time
45 min

Materials
Paper and a pen. No device needed.

Learning goals

- IF this, THEN that
- When should it switch?
- A device that remembers
- Power from batteries and the sun

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Worksheet

Make it automatic: unit check

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Name _____ Class _____ Date _____

1 **Which rule runs a classroom fan only when it's needed?**

- ◯ IF above 30 °C OR a class is in, THEN fan on
- ◯ IF above 30 °C, THEN fan on
- ◯ IF above 30 °C AND a class is in, THEN fan on

2 **A smoke alarm should be set so it almost never gives false alarms, even if it sometimes misses a fire.**

- ◯ True
- ◯ False

3 **Put the steps for choosing an alarm's threshold in order.**

- ☐ Adjust it, and write down why
- ☐ Watch for a few weeks, counting false alarms and misses
- ☐ Put the threshold in the gap between the two
- ☐ Find out what the readings look like when something goes wrong
- ☐ Log readings on normal days

4 **A tank pump is set to switch on below 20% and off above 90%. The tank is at 50% and the pump is running. What happens next?**

- ◯ It keeps running until the tank passes 90%
- ◯ It stops, because 50% is above 20%
- ◯ It clicks on and off around 50%

5 **Match each word to what it means for a sensor's readings.**

Noise	• • • •	The middle reading: one glitch barely moves it
Glitch	• • • •	Add and divide: calmer, but slower to react
Average	• • • •	One reading that can't be right, like 85 °C at dawn
Median	• • • •	The small wobble in every reading